

Difference in differences

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Recap: What we have learned about RCTs

The Power of RCTs

- RCTs are the gold standard for causal inference.
- Randomization eliminates selection bias and heterogeneous treatment effect bias.

However, randomization is not always applicable

- Ethical and Practical Constraints: Some interventions can't be randomized (e.g., policy changes).
- External Validity: RCT results may not generalize to other contexts.
- Cost and Time: RCTs can be expensive and slow to implement.

The Bonus Incentive Case

So when there is a treatment that cannot be randomized, how can we get the causal relationship?

Let's re-consider the bonus incentive case in the retail chain:

- If due to some policy restrictions, the bonus incentive cannot be randomly assigned.
- The bonus incentive is the treatment and is denoted as D , and let the log of sales $\ln(sales_{it})$ be the dependent variable.
- We want to identify the causal effect of D on $\ln(sales_{it})$ when the treatment is not randomly assigned.

The Bonus Incentive Case

The first step, let's try to figure out any potential effects.

- There are some fixed effects, which are unobserved, time-invariant, and unique to each group. We use FE_T and FE_C to denote the fixed effects on treatment and control groups.
- There are natural changes in sales over time, denoted as T .
- There is a treatment effect on the treated group, denoted as D .

The Bonus Incentive Case

- Then immediately we can compare the difference **between the two groups**:

Shops	Outcome
Treated Shops	$FE_T + T + D$
Untreated Shops	$FE_C + T$

- Now let's make a simple difference across the two groups!
$$SDO = D + (FE_T - FE_C)$$
- The second term is something we are familiar with, the selection bias ($E[Y^1|D=0] - E[Y^0|D=0]$)
- Then, shall we consider comparing a group to itself?**

The Bonus Incentive Case

We can use a table to illustrate the sales of the two groups before and after the treatment.

Shops		Outcome
Treated Shops	Before	$FE_T + T + D$
	After	FE_T
Untreated Shops	Before	FE_C
	After	$FE_C + T$

Simply making difference between before and after for each group, we can get:

- The change of the treated group: $T + D$;
- The change of the untreated group: T ;
- The natural changes T cannot be observed and estimated!

The Bonus Incentive Case

But we are almost there..... Why don't we combine the two differences? And that's how Difference in Difference comes!

Shops		Outcome	$Diff_1$	$Diff_1$
Treated Shops	Before	$FE_T + T + D$	T + D	D
	After	FE_T		
Untreated Shops	Before	FE_C	T	
	After	$FE_C + T$		

- The intuition of the DD strategy is remarkably simple: combine these two simpler difference so the selection bias and the effect of time are, in turn, eliminated.

Difference in Difference

In summary, what does DID do?

- The first difference, $Diff_1$, makes the simple before-and-after difference.
- Then, once those differences are made, we **difference the differences** (hence the name) to get the unbiased estimate of D .

Have you found the important assumption we implicitly made?

- We are assuming that there are **no time-variant group-specific unobservables**, equivalent to assuming that T is the same for all units.
- And we call this **the parallel trends assumption**.

The Parallel Trends Assumption

Let's take this assumption seriously. We write the DID estimator $\hat{\delta}$ in the mathematical form:

$$\hat{\delta} = (E[Y_T | \text{Post}] - E[Y_T | \text{Pre}]) - (E[Y_U | \text{Post}] - E[Y_U | \text{Pre}]) \quad (1)$$

Let's use the switching equation and do a little trick to the above equation:

$$\begin{aligned} \hat{\delta} = & \underbrace{\left(E[Y_T^1 | \text{Post}] - E[Y_T^0 | \text{Pre}] \right) - \left(E[Y_U^0 | \text{Post}] - E[Y_U^0 | \text{Pre}] \right)}_{\text{Switching equation}} \\ & + \underbrace{E[Y_T^0 | \text{Post}] - E[Y_T^0 | \text{Post}]}_{\text{Adding zero}} \end{aligned} \quad (2)$$

The Parallel Trends Assumption

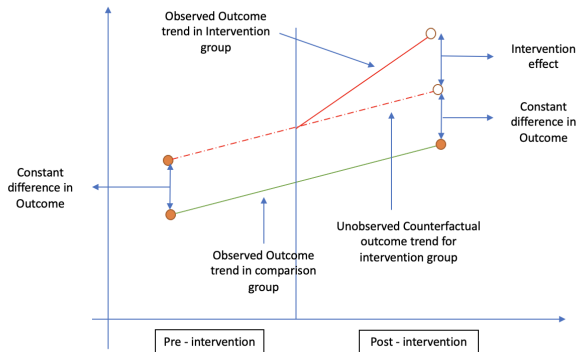
Now we simply rearrange these terms:

$$\begin{aligned}\hat{\delta} = & \underbrace{E[Y_T^1 | \text{Post}] - E[Y_T^0 | \text{Post}]}_{\text{ATT}} \\ & + \underbrace{\left[E[Y_T^0 | \text{Post}] - E[Y_T^0 | \text{Pre}] \right] - \left[E[Y_U^0 | \text{Post}] - E[Y_U^0 | \text{Pre}] \right]}_{\text{Non-parallel trends bias}}\end{aligned}\quad (3)$$

- When the non-parallel trends bias becomes 0, the DID estimator will be an unbiased estimator.
- This is exactly what we have assumed, **the parallel trends assumption**.
- What's the intuition behind this assumption?

The Parallel Trends Assumption

By assuming the non-parallel trends, we are actually assuming:



- Units not randomly assigned $\Rightarrow$ selection bias $\neq 0$.
- But changes are as good as randomly assigned $\Rightarrow$ $E[\text{change of selection bias}] = 0$

Requirements for a "good treatment shock"

Then how can we ensure this assumption is true?

We need some good treatment shocks.

What makes a good treatment shock?

- **Shock strength:** Strong enough to significantly change units's behavior.
- **Exogenous shock:** Came from outside the system.
 - ▶ Units did not choose to be treated, could not anticipate the shock, no reason to think unobservable predict potential outcomes or which units were treated.

Requirements for a "good treatment shock"

What makes a good treatment shock? (Continued)

- **Covariate balance:** Reasonable covariate balance between treated and control units, including a common support.
 - ▶ Somewhat imperfect balance can be addressed with balancing methods.
- **Only-through condition(s):** The effect of the shock on the outcome must come only through the shock.
 - ▶ No other shock, at around the same time, could affect treated units differently than control units.

Test The Key Assumption

Do we have any method to test if the parallel trends assumption is satisfied?

- Unfortunately, the assumption cannot be directly observed and verified.
 - ▶ $E[Y_T^0|Post]$ is counterfactual and is not observable.
- Instead, we usually compare placebo pre-treatment leads of the DID coefficient (**The Pre-treatment Balance**).
 - ▶ If DID coefficients in the pre-treatment periods are statistically zero, then the difference-in-differences between treatment and control groups followed a similar trend prior to treatment.
- And here's the rhetorical art of the design: if they had been similar before, then why wouldn't they continue to be post-treatment?
 - ▶ However, be careful about the fallacy: reverse position.

Checking The Pre-treatment Balance

Consider different scenarios.

- When a set of groups who received the treatment at the same point in time,
 - ▶ Simply show the raw data, and visually inspect whether the pre-treatment dynamics of the treatment group differed from that of the control group units.
- When groups of units adopt the treatment at different points,
 - ▶ One possibility is to plot the raw data, year by year, and simply eyeball.
 - ▶ Anderson, Hansen, and Rees (2013)¹.
 - ▶ Estimate a regression model that includes treatment leads and lags.

¹<https://doi.org/10.1086/668812>

Summary of How to Conduct a DID Analysis

- 1 Specify your hypotheses. For example,
 - ▶ Null hypothesis: There is no causal impact from the treatment.
 - ▶ Alternative hypothesis: There is a causal impact from the treatment.
- 2 Define the treatment group and the control group.
- 3 Learn about the data.
- 4 Test if the parallel trends assumption is satisfied.
- 5 Run a regression.
- 6 Interpret the results and get managerial implications.

Let's Use DID now!

Continue with the bonus incentive scheme example and suppose we have collected all the data we need.

First, we specify the hypotheses.

- Null: The bonus incentive scheme has no causal impact on sale amounts.
- Alternative: The bonus incentive scheme has either positive or negative impacts on sale amounts (two-sided test).

Let's Use DID now!

Second, let's look at the data. Different from the RCT case, in which we use cross-sectional data, in the DID case, we will deal with panel data.

	Time series data	Cross-sectional data	Panel data
Number of individuals	One individual	Many individuals	Many individuals
Time periods	Across many time periods	At a particular point in time	Across many time periods

- Panel data can be of one of these two types:
 - ▶ Balanced panel data: All the individuals are observed in all time periods.
 - ▶ Unbalanced panel data: All the individuals are not observed in all time periods.

Let's Use DID now!

Here is an example for balanced data:

Balanced Data				
Shop ID	Sales amount	Treatment	Time	
1	16.41	0	2010	
1	10.29	0	2011	
1	16.07	0	2012	
2	18.21	0	2010	
2	19.79	0	2011	
2	13.65	0	2012	
3	18.49	0	2010	
3	12.23	0	2011	
3	18.26	0	2012	
4	14.89	0	2010	
4	16.00	0	2011	
4	11.59	0	2012	

Let's Use DID now!

This is an example for unbalanced data:

Unbalanced Data				
Shop ID	Sales amount	Treatment	Time	
1	16.41	0	2010	
1	10.29	0	2011	
2	18.21	0	2010	
2	19.79	0	2011	
2	13.65	0	2012	
3	18.49	0	2010	
3	12.23	0	2011	
3	18.26	0	2012	
4	14.89	0	2010	
4	16.00	0	2011	

Let's Use DID now!

Any difference between using balanced data and unbalanced data?

- Balanced data generally provide more consistent and reliable DID estimates.
- Unbalanced data can introduce bias into the DID estimator.
 - ▶ If the missing data or unobserved periods are systematically related to the treatment or outcome, the estimated treatment effect might be distorted.
 - ▶ If the missing data in an unbalanced dataset is random, the bias might be minimal.
- It's crucial to understand why the data is unbalanced and consider strategies to mitigate potential biases
 - ▶ Conduct robustness checks, such as using imputation methods or sensitivity analysis.

Let's Use DID now!

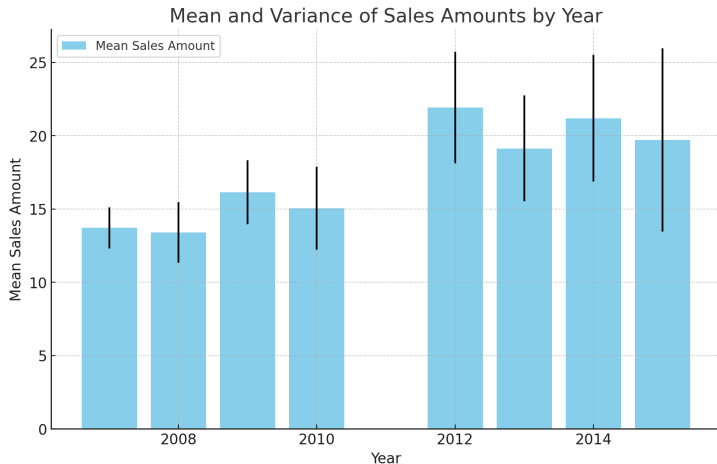
In our example, the data is balanced. and let's ask ChatGPT to help us do statistical descriptions.

Time	Mean	Variance
2007	13.71	1.95
2008	13.40	4.25
2009	16.13	4.74
2010	15.05	7.97
2012	21.92	14.47
2013	19.12	13.02
2014	21.19	18.64
2015	19.71	39.05

Table: Mean and Variance of Sales by Year

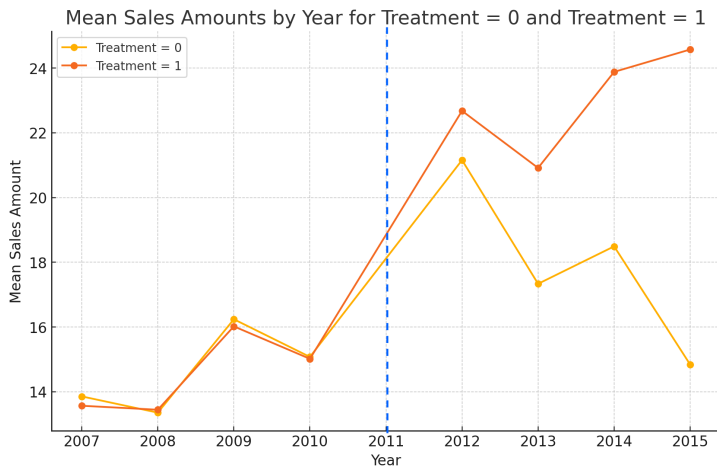
Let's Use DID now!

A figure seems to be better.



Let's Use DID now!

Then let's examine the pre-treatment trends. Do you think we can make the parallel trends assumption?



Let's Use DID now!

We can do DID regression now. The regression function is:

$$\text{Sales}_{it} = \beta_0 + \beta_1 \cdot \text{Treatment}_i + \beta_2 \cdot \text{Post}_t + \beta_3 \cdot (\text{Treatment}_i \times \text{Post}_t) + \epsilon_{it} \quad (4)$$

Table: DID Regression Results

	Coef.	Std.Err.	t	P> t	[0.025	0.975]
Intercept	14.63	0.23	62.50	0.00	14.17	15.09
Treatment	-0.12	0.33	-0.36	0.72	-0.77	0.53
Post	3.32	0.33	10.04	0.00	2.67	3.97
Treatment_Post	5.18	0.47	11.05	0.00	4.26	6.09

The last row is of our interest.

Let's Use DID now!

DID Regression in Python (Part 1)

```
import pandas as pd
import statsmodels.formula.api as smf

# Assuming the data is already loaded into a DataFrame named '
    data'
# If you haven't loaded the data yet, you can load it like this
:
# data = pd.read_excel('your_file_path.xlsx', sheet_name='
    Sheet1')

# Create the Post variable, indicating whether the observation
    is in the post-treatment period
data['Post'] = (data['Time'] > 2010).astype(int) # Modify
    '2010' if the treatment year is different
```

Listing 1: Part 1

Let's Use DID now!

DID Regression in Python (Part 2)

```
# Create the interaction term for DID
data['Treatment_Post'] = data['Treatment'] * data['Post']

# Run the DID regression
model = smf.ols('Sales_amount ~ Treatment + Post +
               Treatment_Post', data=data).fit()

# Display the regression results
print(model.summary())
```

Listing 2: Part 2

Let's Use DID now!

How do you interpret the results?

- The coefficient of interest is β_3 .
- The positive and statistically significant value (5.18, $p < 0.001$) indicates that the incentive scheme had a significant positive impact on sales for the treatment group.
- Specifically, the incentive scheme increased sales by an additional 5.18 units in the treatment group compared to the control group.

Let's Use DID now!

What managerial implications can we learn from it?

- Effectiveness of the Incentive Scheme.
- Focus on Further Enhancements.
- Tailoring the Strategy according to some heterogeneous effects tests (Not done in this example).

Classical Example: Card & Krueger Research

Research question: Do (modestly) higher minimum wages reduce low-wage employment?

Card and Krueger considered the impact of New Jersey's 1992 minimum wage increase from 4.25 to 5.05 per hour.

- In terms \$ in 2013: Equivalent to an Increase from \$7.00 to \$8.30
- Compare US minimum wage \$7.25; Illinois minimum wage: \$8.25

Card and Krueger compared 410 fast-food restaurants in **New Jersey (treated)** to eastern **Pennsylvania (control)** before and after the increase.

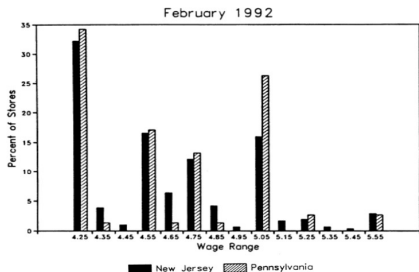
Classical Example: Card & Krueger Research

Motivating example

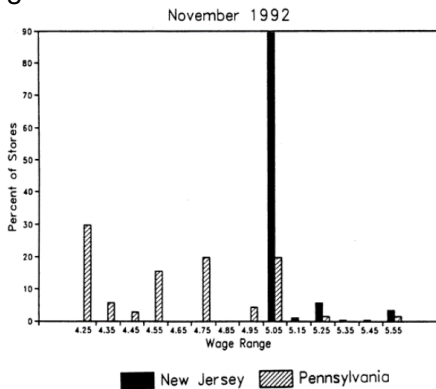
- Data on wages and employment:
 - ▶ March & Dec 1992: one month before, 8 months after the increase.
- Note the "local" nature of the question:
 - ▶ Microeconomic theory: raise minimum enough $\Rightarrow$ lower employment.
 - ▶ Higher prices $\Rightarrow$ lower equilibrium demand.
 - ▶ Over time, higher labor cost $\Rightarrow$ substitute capital for labor.

Classical Example: Card & Krueger Research

Wages before and after minimum wage increase



Wages Before Treatment



Wages After Treatment

Classical Example: Card & Krueger Research

Selected Card & Krueger results

- Full-time equivalent employment, per restaurant.
- Drop 6 restaurants which closed from pre to post; 4 which temporarily closed.
- NJ looks better after minimum wage increase:

Time	PA	NJ	NJ - PA
Before	23.33	20.44	-2.89
	(1.35)	(0.51)	(1.44)
After	21.17	21.03	-0.14
	(0.94)	(0.52)	(1.07)
After - Before	-2.16	0.59	2.76**
	(1.25)	(0.54)	(1.36)

- But the effect is entirely because PA employment declines.
- Puts great stress on the parallel changes assumption.

Classical Example: Card & Krueger Research

Regression implementation of DiD

- Time $t=0$ (before) or 1 (after)
- w_i = dummy for treatment group

$$y_{it} = \alpha + \gamma \cdot w_i + \beta \cdot t + \delta_{DiD} \cdot w_i \cdot t + \epsilon_{it} \quad (5)$$

- Always include non-interacted terms. Why? If we drop $\gamma \cdot w_i$ what happens?
- It will be absorbed into the error term.

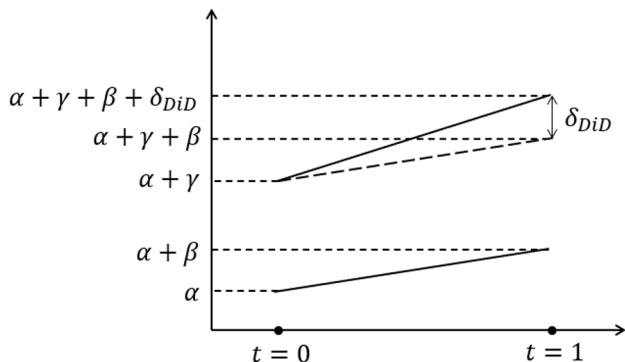
$$\begin{aligned} \text{Corr}(w_i, w_i \cdot t) &\neq 0 \quad (\text{often large}) \\ \Rightarrow \text{Corr}(\epsilon_i, w_i \cdot t) &\neq 0 \quad (\text{often large}) \\ &\Rightarrow \text{omitted variable bias} \end{aligned}$$

- δ_{DiD} will capture some of the impact of (omitted) w_i

Classical Example: Card & Krueger Research

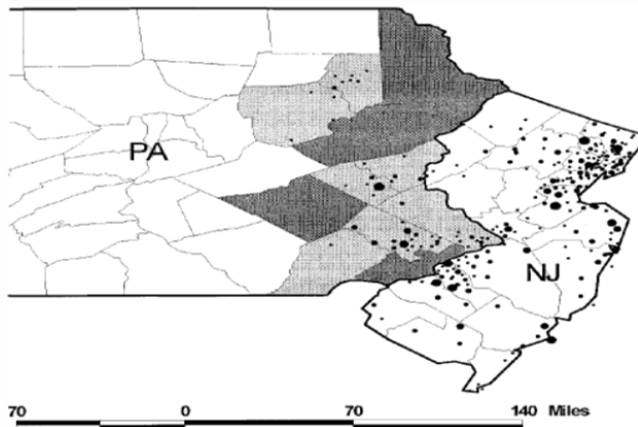
Meaning of coefficients on interaction terms

$$y_{it} = \alpha + \gamma \cdot w_i + \beta \cdot t + \delta_{DiD} \cdot w_i \cdot t + \epsilon_{it}$$



Classical Example: Card & Krueger Research

Are these locations similar enough?



Classical Example: Card & Krueger Research

DiD threat: Non-parallel changes

- How can we test for parallel changes?
- Basic strategy: Use multiple pre-periods
 - ▶ See if (visually) parallel changes over $t = [-n, 0]$.
 - ▶ If not parallel, assumption not justified over $t = [0, 1]$.

Classical Example: Card & Krueger Research

Longer trends in NJ v. PA fast-food employment

- One surely wouldn't conclude that NJ employment rose.
- Better research design: pre-period data for a longer time.
 - ▶ Data not available.
- Still, no strong evidence of NJ employment drop

